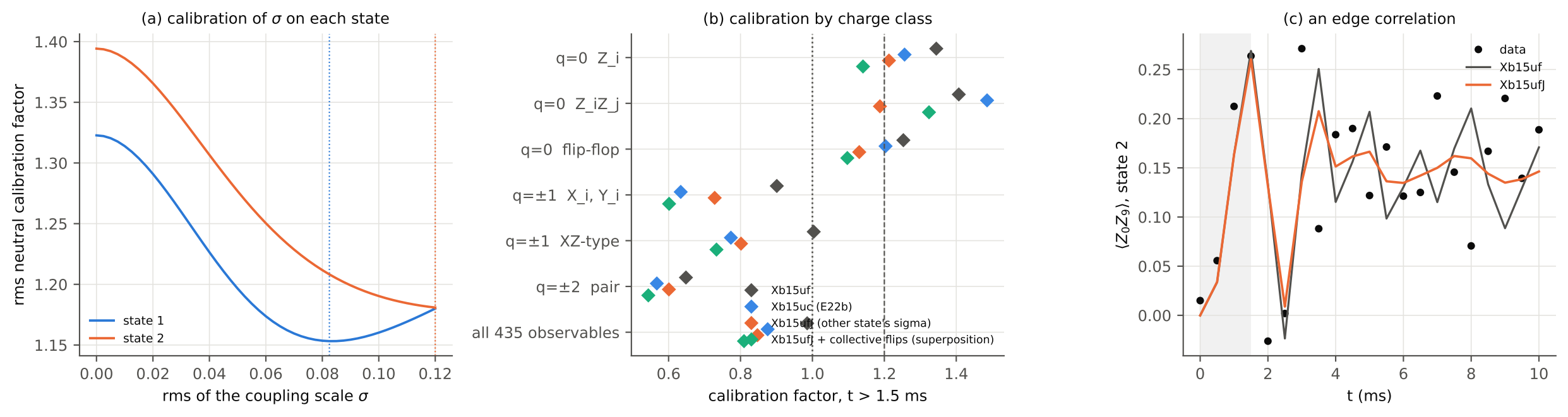
(a) calibration of σ on each state
rms neutral calibration factor
1.40
1.35
1.30
1.25
1.20
1.15
state 1
state 2
0.00
0.02
0.04
0.06
0.08
0.10
0.12
rms of the coupling scale σ
(b) calibration by charge class
q=0 Z_i
q=0 Z_iZ_j
q=0 flip-flop
q=±1 X_i, Y_i
q=±1 XZ-type
q=±2 pair
all 435 observables
Xb15uf
Xb15uc (E22b)
Xb15uf (other state's sigma)
Xb15ufJ + collective flips (superposition)
0.6
0.8
1.0
1.2
1.4
calibration factor, t > 1.5 ms
(c) an edge correlation
⟨Z0Z9⟩, state 2
0.25
0.20
0.15
0.10
0.05
0.00
data
Xb15uf
Xb15ufJ
0
2
4
6
8
10
t (ms)